

Model Miniaturization for Medical Triage: A Comparative Study of Compression and Fine-Tuning Methods

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Abstract—Local, private, and offline deployment of Large Language Models (LLMs) in clinical settings is highly desirable but constrained by the hardware and memory requirements of state-of-the-art architectures. This report presents a systematic, reproducible comparative study of model compression and parameter-efficient fine-tuning methods for building a lightweight, resource-constrained medical triage assistant. We investigate structural pruning, logit-level knowledge distillation, and Low-Rank Adaptation (LoRA) fine-tuning of an 8-billion-parameter medical LLM (OpenBioLLM-Llama3-8B) into a 0.6-billion-parameter student (Qwen3-0.6B). All methods are evaluated on a unified harness spanning four clinical triage datasets, from fully synthetic to real, de-identified hospital records (MIMIC-IV-ED). Direct LoRA fine-tuning on a mixture of teacher-generated and real clinical data achieves the strongest overall performance (90.8% accuracy on the combined suite; 80.2% accuracy and 90.4% emergency recall on MIMIC-IV-ED), while attention-head pruning with recovery fine-tuning attains comparable accuracy (90.5%) with an 18% reduction in network depth. Knowledge distillation proves the most fragile technique, suffering two distinct training collapses—traced to a numerical overflow bug and a class-imbalance-driven prediction bias—before stabilizing under a frozen-base, LoRA-adaptor recovery scheme. All compressed models occupy under 1 GB of VRAM and answer in a median of 0.91 seconds per sample, a 13.3× reduction in parameters and 10.6× reduction in VRAM relative to the teacher, while matching or exceeding the teacher’s accuracy on every held-out set. We additionally uncover a vocabulary-leakage vulnerability in a commonly used real-patient domain-shift benchmark and argue for MIMIC-IV-ED as the primary hard benchmark for comparative clinical triage studies.

Index Terms—model compression, knowledge distillation, structured pruning, LoRA, parameter-efficient fine-tuning, medical triage, edge deployment, large language models.

I. INTRODUCTION

Medical triage is the process of sorting patients in an emergency department (ED) by the severity of their condition, in order to prioritize scarce clinical attention and minimize the risk that a life-threatening case is left waiting. Historically, triage decisions have been made by trained nurses using structured protocols such as the Emergency Severity Index (ESI), a five-level scheme ranging from Level 1 (immediate, life-saving intervention required) to Level 5 (non-urgent) [1]. Large Language Models (LLMs) have recently demonstrated a

strong ability to read free-text clinical complaints and produce structured triage decisions together with a supporting rationale, raising the prospect of AI-assisted triage support at the point of care.

Deploying such models in a real clinical environment, however, introduces constraints that are largely absent from general-purpose chatbot use cases. Sending patient data to a third-party cloud API raises immediate concerns around privacy, regulatory compliance (HIPAA, GDPR), and availability during network outages. The alternative—running the model locally on modest hospital or bedside hardware—is attractive but is blocked by the sheer size of state-of-the-art medical LLMs. The teacher model used in this work, OpenBioLLM-Llama3-8B, requires approximately 5.7 GB of VRAM even in an aggressively compressed 4-bit format, and well over 16 GB in native precision.

This report investigates **model miniaturization**: the set of techniques used to compress the capabilities of a large teacher model into a small, fast, cheap-to-run student model, while retaining as much of the teacher’s task performance as possible. We compress OpenBioLLM-Llama3-8B (8.03B parameters) into Qwen3-0.6B (0.62B parameters)—a 13.3× reduction in parameter count—and evaluate how well three miniaturization strategies preserve clinical triage accuracy and, critically, the recall of true emergencies.

A. Motivation for Miniaturization

The motivation for aggressive miniaturization rests on four practical pillars that are especially acute in a clinical setting: (i) **Memory footprint**—consumer and hospital-grade edge hardware rarely exceeds a few gigabytes of usable VRAM, and a model that cannot fit in memory cannot be deployed at all; (ii) **Latency**—triage is time-critical, and a clinician cannot wait several seconds for a cloud round-trip; (iii) **Energy and cost efficiency**—smaller models require less compute per inference, lowering electricity and infrastructure cost for under-resourced sites; and (iv) **Edge and offline deployment**—rural clinics, ambulances, and disaster-response settings may have no reliable connectivity, so a model that only works online is not a viable clinical tool.

B. Project Goals

We set four measurable targets for the compressed student: (1) memory footprint under 1 GB of VRAM, ideally near 500 MB; (2) triage accuracy of at least 85% on in-domain and external clinical data; (3) clinical safety (emergency recall) of at least 95%, and ideally 100%, since misclassifying a true emergency as routine carries a severe, asymmetric clinical risk relative to a false alarm; and (4) inference latency under 1.0 second per sample to support real-time use.

C. Triage Class Definitions

To keep the target simple and clinically interpretable, we compress the five-level ESI scale into three classes: **EMERGENCY** (ESI 1–2): immediately life-threatening, intervention within minutes (e.g. myocardial infarction, stroke, anaphylaxis); **URGENT** (ESI 3): serious but stable, evaluation within 1–2 hours (e.g. moderate abdominal pain, high fever without sepsis signs); and **ROUTINE** (ESI 4–5): non-urgent, safely schedulable (e.g. minor abrasions, common colds, prescription refills).

D. Contributions

This paper makes four contributions. First, it runs a *controlled, head-to-head comparison* of three structurally different compression strategies on a single clinical task under an identical, failure-mode-aware evaluation harness. Second, it provides a fully worked mathematical and procedural specification of label-space-restricted logit distillation, including a numerical-precision failure mode and its resolution. Third, it identifies and documents a vocabulary-leakage vulnerability in a commonly used real-patient benchmark. Fourth, it delivers an open, modular, reproducible codebase in which every method records both a zero-shot baseline and a tuned result. Section II reviews related work; Section III formalizes the problem; Section IV details the methods; Section V describes the codebase; Section VI reports results; Sections VII–IX discuss trade-offs, limitations, and ethics; and Section X concludes. Per-author contributions appear in Section XI.

II. BACKGROUND AND RELATED WORK

A. Model Compression for Large Language Models

Four broad families of technique dominate practical LLM compression, and this project draws on all four. **Pruning** removes redundant parameters, heads, or layers from a trained network. Michel et al. [2] showed that a large fraction of attention heads can be removed at inference with minimal accuracy loss, motivating importance-score-based head pruning. Frantar and Alistarh [3] demonstrate that even very large models can be pruned to high sparsity in one shot using approximate second-order information; our routine (Section IV-C) uses a first-order activation proxy for tractability on a single-GPU budget. **Quantization** reduces numerical precision, trading fidelity for footprint. We use 4-bit NormalFloat (NF4) via `bitsandbytes`, introduced with QLoRA by Dettmers et al. [4], both to load the 8B teacher affordably and to compress the deployed student. **Knowledge distillation (KD)** trains a

small student to reproduce a larger teacher’s output distribution [5], and has been applied to compress Transformers, e.g. DistilBERT [6] and TinyBERT [7]. Section IV-D adapts this to a closed three-label output space. **Low-Rank Adaptation (LoRA)** [8] freezes pretrained weights and injects small trainable low-rank updates, cutting trainable parameters sharply while retaining most of full fine-tuning’s performance. We use LoRA both standalone and as a post-pruning recovery mechanism.

These four families are not mutually exclusive, and much of the practical value of modern compression comes from *composing* them: quantization reduces the footprint of a model that has already been pruned or distilled, and LoRA recovery repairs the capability lost to a lossy compression step without the cost of full fine-tuning. The QLoRA recipe [4], which fine-tunes low-rank adapters on top of a 4-bit-quantized frozen base, is the canonical example and is used throughout this project. What is comparatively under-explored in the literature—and what this study targets—is a controlled comparison of these compositions on a single *safety-critical* task, where the relevant question is not only average accuracy but whether the rare, high-cost class survives each compression step intact.

B. Clinical NLP and Triage Automation

Assisting emergency triage with learned models is an active area of clinical informatics, motivated by significant inter-rater variability in human ESI assignment. The MIMIC-IV-ED database [9], a large de-identified ED dataset, is a standard resource and serves as our primary external validation set. The base models used—Qwen3 [10] as student and a Llama-3 [11]-derived biomedical model as teacher—both post-date instruction tuning, which makes zero-shot prompting for structured triage a reasonable (though, as Section VI-B shows, very weak) starting point.

Inter-rater variability is worth dwelling on, because it shapes what “accuracy” can even mean for this task. Two experienced nurses presented with the same borderline case do not always assign the same ESI level, so the ground-truth labels against which any automated system is scored are themselves noisy. This has two consequences for our study. First, it bounds the achievable accuracy of a perfectly calibrated model from above: some fraction of apparent errors reflect label disagreement rather than model failure. Second, and more constructively, it creates a genuine role for a consistent automated second reader, whose value lies less in being “more correct” than a nurse and more in being reproducible and available at the point of care—provided it never silently down-triages a true emergency, which is exactly why we foreground emergency recall over raw accuracy throughout.

C. Positioning of This Work

Much of the compression literature reports a single method on a single general benchmark. This project instead runs a *controlled, head-to-head comparison* of three compression philosophies on one clinical task, using identical splits and

identical failure-mode accounting (parsing failures counted separately from misclassifications). We view this comparative, failure-mode-aware protocol, together with the benchmark-vulnerability finding of Section VI-D, as at least as valuable as any single trained artifact.

III. PROBLEM DEFINITION AND OBJECTIVES

A. Formal Problem Statement

Let x denote a free-text patient presentation and $y \in C = \{\text{EMERGENCY}, \text{URGENT}, \text{ROUTINE}\}$ the ground-truth class. Given a teacher f_T with parameters $\theta_T \in \mathbb{R}^{8.03B}$, we seek a student f_S with $\theta_S \in \mathbb{R}^{0.62B}$ minimizing the expected task loss subject to a memory constraint:

$$\theta_S^* = \arg \min_{\theta_S} \mathbb{E}_{(x,y) \sim \mathcal{D}} [\mathcal{L}(f_S(x; \theta_S), y, f_T(x; \theta_T))], \quad (1)$$

subject to $\text{VRAM}(\theta_S) < 1 \text{ GB}$. Here $\mathcal{L}$ is a task loss that may use the teacher’s predictions directly (distillation), indirectly via teacher-labeled training data (LoRA), or not at all beyond the pretrained initialization (pruning), and $\mathcal{D}$ is the joint distribution over the four evaluation datasets of Section VI-A.

B. Optimization Targets and Constraints

Optimizing (1) alone is insufficient for a safety-critical task: an on-average-accurate student may still be dangerous if its errors concentrate on the EMERGENCY class. We therefore decompose the objective into the four individually-tracked targets of Section I-B: $\text{VRAM} < 1 \text{ GB}$, $\text{accuracy} \geq 85\%$, $\text{emergency recall} \geq 95\%$, and $\text{latency} < 1.0 \text{ s/sample}$. A student is acceptable only if it satisfies the VRAM and latency constraints *and* clears the accuracy and recall bars.

C. Two Competing Strategies

We compare two independently-developed approaches. **Approach 1 (Compression-centric)**: physically shrink the architecture via structured head pruning and layer dropping, and recover capability via knowledge distillation. **Approach 2 (Data-centric)**: leave the architecture untouched and instead invest in a larger, higher-quality training signal—teacher-generated synthetic reasoning combined with real clinical records—fine-tuned in with LoRA. Both start from the same pretrained Qwen3-0.6B checkpoint and are scored on the same held-out data, enabling a fair comparison.

IV. METHODOLOGY

Figure 1 shows the full pipeline. Both approaches begin from the same pretrained Qwen3-0.6B checkpoint and the same teacher-generated data infrastructure, and diverge only in how that starting point is transformed into a deployable model.

A. System Architecture

Table I summarizes the two base models. The teacher is loaded in 4-bit NF4 precision throughout, keeping its footprint at 5.7 GB and allowing teacher and student to reside on a single GPU during distillation.

Algorithm 1: Structured Head and Layer Pruning

Input: Pretrained student f_S , calibration batch $\mathcal{B}$, head-prune ratio $\rho=0.4$, layers to drop $k=5$
Output: Pruned student $f_S(\cdot; \theta'_S)$
1 Forward f_S over $\mathcal{B}$; cache attention maps $A_{b,h,i,j}$;
2 **for each layer** l **do**
3 **for each head** h **do**
4 Compute $I(l, h)$ via Eq. (2);
5 Rank ascending; zero W_q, W_k, W_v slices of the bottom $\rho \cdot 16$ heads;
6 Compute $\bar{I}(l) = \text{mean}_h I(l, h)$; drop the k contiguous middle layers of lowest $\bar{I}(l)$;
7 **return** θ'_S

B. Synthetic Data Generation

Manually annotating a large triage dataset is costly, so the teacher bootstraps training data via **oracle-label prompting**: rather than triaging from scratch (which biases toward the cautious EMERGENCY/URGENT classes), the teacher is supplied a ground-truth label and a seed description and asked to generate the consistent clinical reasoning. A structural response primer (TRIAGE LEVEL: {label} / KEY SYMPTOMS:) forces immediate structured output rather than conversational filler. This yielded 5,100 synthetic records, split 80/10/10 into 4,080 train, 510 validation, and 510 test. As a semantic sanity check, roberta-large BERTScore against PubMedQA reference prose gave an overall F1 of **0.8196** (0.8191/0.8193/0.8202 for the three classes), indicating consistent coherence rather than class-skewed generation.

C. Approach 1: Structured Attention-Head Pruning

Attention heads in a trained Transformer are frequently redundant [2]. We compute an importance score for every head using a mean-squared activation proxy over a calibration batch:

$$I(l, h) = \frac{1}{B S^2} \sum_{b=1}^B \sum_{i=1}^S \sum_{j=1}^S A_{b,h,i,j}^2, \quad (2)$$

where l indexes the layer, h the head, $B=128$ the batch size, $S=256$ the sequence length, and $A_{b,h,i,j}$ the attention weight from query i to key j . Within each layer, heads are ranked by $I(l, h)$ and the lowest-scoring 40% are pruned by zeroing their projection slices, $W_q[:, s:e] = W_k[:, s:e] = W_v[:, s:e] = 0$, with $s = h d_{\text{head}}$, $e = (h+1) d_{\text{head}}$, and $d_{\text{head}}=56$. This is a *soft* form of pruning: tensors are masked rather than resized, which preserves the model configuration at the cost of not reducing on-disk size or compute. Separately, using per-layer average importance, we drop the five contiguous middle layers with lowest aggregate activation variance, $\mathcal{P}_{\text{drop}} = \{10, 12, 14, 16, 18\}$, reducing effective depth from 28 to 23 layers ($\approx 18\%$). Algorithm 1 summarizes the procedure.

D. Approach 1: Knowledge Distillation

Following pruning, we apply logit-level KD from the teacher. Our task requires only a single label token from a closed set C ; distilling over the full vocabulary is ill-posed since teacher and student tokenizers differ (128k vs. 151k).

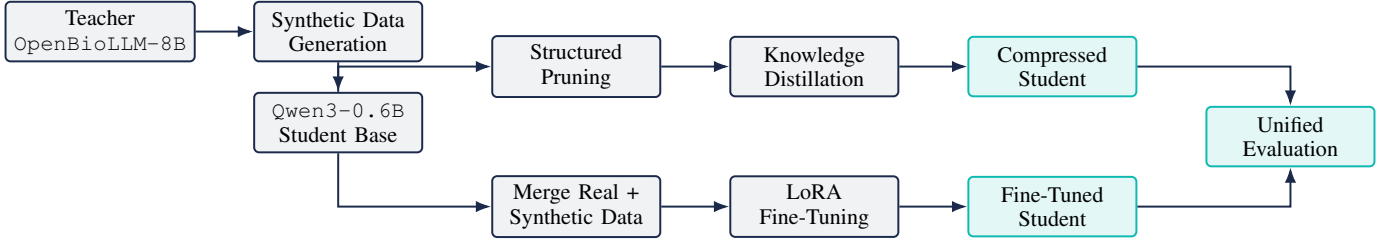


Fig. 1. End-to-end pipeline. The teacher generates synthetic training data, which feeds both the compression-centric (top) and data-centric (bottom) branches; both final student variants are scored on an identical evaluation harness. What the reader should see is one shared starting point diverging into two strategies and reconverging at a common evaluation, and what it means is that every reported difference is attributable to the method, not the setup.

TABLE I
MODEL ARCHITECTURE SPECIFICATIONS. TEACHER LOADED IN 4-BIT NF4; STUDENT FOOTPRINT REPORTED IN FP16.

Role	Model	Parameters	Layers	Heads	VRAM	Precision
Teacher	OpenBioLLM-Llama3-8B	8.03 B	32	32	5.7 GB	4-bit NF4
Student	Qwen3-0.6B	0.62 B	28	16	0.54 GB	fp16

We map each model’s next-token logits onto C by averaging the logits of the sub-word tokens realizing each label word:

$$z_c = \frac{1}{|V(c)|} \sum_{\text{id} \in V(c)} \text{logit}_{\text{id}}, \quad c \in C, \quad (3)$$

where $V(c)$ is the token-id set realizing label c . Temperature-scaled soft distributions are then computed as

$$p_c^m = \frac{\exp(z_c^m / T_{\text{kd}})}{\sum_{j \in C} \exp(z_j^m / T_{\text{kd}})}, \quad m \in \{T, S\}, \quad (4)$$

applied identically to the teacher ($m=T$) and student ($m=S$), and the student matches the teacher via temperature-scaled KL divergence, combined with cross-entropy on the true label:

$$\mathcal{L}_{\text{KD}} = T_{\text{kd}}^2 \sum_{c \in C} p_c^T \log \left(\frac{p_c^T}{p_c^S} \right), \quad (5)$$

$$\mathcal{L}_{\text{total}} = \alpha_{\text{kd}} \mathcal{L}_{\text{KD}} + (1 - \alpha_{\text{kd}}) \mathcal{L}_{\text{CE}}(p^S, y). \quad (6)$$

The first configuration ($\alpha_{\text{kd}}=0.7$, $T_{\text{kd}}=4.0$, full-model) showed a mild bias toward URGENT under greedy decoding. A second, more aggressive configuration ($\alpha_{\text{kd}}=0.5$, $T_{\text{kd}}=2.0$, EMERGENCY class weight 3.0) instead triggered two compounding failures: (i) a NaN loss explosion, traced not to the student but to the *teacher*—its 4-bit float16 attention scores occasionally exceeded float16’s range ($\approx 65,504$) and overflowed to Inf, corrupting the KD targets—fixed by forcing the teacher’s compute dtype to bfloat16; and (ii) even after the fix, unrestricted full-parameter fine-tuning against the heavily weighted loss over-corrected into a majority-class collapse. Only restricting distillation to a *frozen* pruned base with a small trainable LoRA adapter stabilized training (Fig. 5), at the cost of a residual bias toward EMERGENCY (Section VIII). Algorithm 2 gives the stabilized per-step training loop.

E. Baseline: TF-IDF + Logistic Regression

As a deliberately simple non-neural reference point, we train a bag-of-words TF-IDF vectorizer followed by a multinomial

Algorithm 2: Logit-Level Knowledge Distillation Step

Input: Frozen teacher f_T (bfloat16), pruned student base with LoRA adapter $f_S(\cdot; \theta_S, \phi)$, batch $\{(x_i, y_i)\}_{i=1}^n$, temperature T_{kd} , mixing weight α_{kd}
Output: Updated adapter parameters ϕ

- 1 **for each** (x_i, y_i) **in batch do**
- 2 Teacher next-token logits $\rightarrow z^T$ via Eq. (3);
- 3 Student next-token logits $\rightarrow z^S$ via Eq. (3);
- 4 Soft distributions p^T, p^S via Eq. (4);
- 5 Accumulate $\mathcal{L}_{\text{KD}}$ (Eq. (5)) and $\mathcal{L}_{\text{CE}}(p^S, y_i)$;
- 6 Batch loss $\mathcal{L}_{\text{total}}$ (Eq. (6)); clip gradients to unit norm;
- 7 Backpropagate through ϕ only (θ_S, θ_T frozen); update ϕ with AdamW;
- 8 **return** ϕ

logistic-regression classifier on the same training split used for the neural methods. This baseline has no notion of word order, no pretrained knowledge, and no clinical reasoning capability; it can only associate surface vocabulary with labels. Its role is twofold: to establish the floor that any neural method must clear to justify its cost, and—as Section VI-D shows—to act as a diagnostic probe for benchmark quality, since a task that a bag-of-words model solves perfectly is, by construction, not testing clinical generalization.

F. Approach 2: Low-Rank Adaptation (LoRA) Fine-Tuning

As a data-centric alternative, we fine-tune the *unpruned* student with LoRA [8], which freezes $W_0 \in \mathbb{R}^{d \times k}$ and represents the update as a low-rank product:

$$W = W_0 + \frac{\alpha_{\text{loa}}}{r} BA, \quad (7)$$

with $B \in \mathbb{R}^{d \times r}$, $A \in \mathbb{R}^{r \times k}$, and $\text{rank } r \ll \min(d, k)$, so that only A and B are updated during training. We use $r=16$, $\alpha_{\text{loa}}=32$ on all attention and MLP projections, under 4-bit NF4 base quantization (QLoRA-style [4]). NF4 quantizes each

weight to one of 16 non-uniform levels chosen for normally-distributed weights,

$$\hat{w} = s \cdot q_i, \quad q_i = Q_{\mathcal{N}(0,1)}\left(\frac{i}{16}\right), \quad i \in \{0, \dots, 15\}, \quad (8)$$

where s is a per-block scale and $Q_{\mathcal{N}(0,1)}$ the standard-normal quantile function; this allocates more precision near zero where transformer weights concentrate.

G. Data Integration for Approach 2

The LoRA set merges the 4,080 synthetic training samples with three real sources—a 500-sample Syntech-500 benchmark, the MIMIC-IV-ED training partition, and the Denmark/Turkey subset of the multi-country fedmml corpus—for 42,872 samples total. An NLI-based quality filter was tried but removed under 4% of samples and was dropped as more complexity than its marginal benefit justified—an explicit case of a technique tried, measured, and discarded. Fine-tuning used effective batch size 32, learning rate 2×10^{-4} (cosine), fp16, 3 epochs.

V. IMPLEMENTATION DETAILS

The project is a modular Python repository with one directory per pipeline stage: `src/data_generation/` (synthetic generation, oracle-label prompting), `src/pruning/` (importance scoring, head zeroing, layer dropping), `src/distillation/` (KD trainers and pruned-base variants), `src/finetuning/` and `src/approach2/` (LoRA training, dataset merging, real-data preprocessing, the logistic-regression baseline, latency profiling), `src/evaluation/` (per-dataset evaluators and the threshold-sweep evaluator), and `src/demo/` (Gradio triage chatbots). Large artifacts (checkpoints, JSONL data) are excluded from version control. Every stage is a standalone, argument-driven script, illustrated below:

```
# 1. Synthetic oracle-labeled data
python src/data_generation/teacher_inference.py
python src/data_generation/prepare_dataset.py
# 2. Approach 1: prune, then distill on frozen base
python src/pruning/headpruning.py --prune_ratio 0.4
python src/pruning/layerdropping.py --drop 10,12,14,16,18
python src/distillation/kdtrainer_pruned_v2.py
# 3. Approach 2: merge datasets, then LoRA fine-tune
python src/approach2/merge_datasets.py
python src/approach2/lora_finetune.py --rank 16 --epochs 3
# 4. Evaluate on the unified harness
python src/evaluation/evaluate_threshold_sweep.py
```

Two engineering practices are worth noting. First, trainer iterations are preserved rather than overwritten (`kdtrainer.py` and `kdtrainer_v2.py` coexist with inline failure-mode comments), which made the debugging narrative of Section IV-D reconstructible. Second, every method records both a zero-shot baseline and a tuned result on the same harness, which is what surfaced the format-collapse failure mode.

VI. EXPERIMENTS AND RESULTS

A. Evaluation Datasets

Models are evaluated on four datasets spanning a difficulty gradient (Table II). The MIMIC-IV-ED demo contains 207 patients; 165 enter the v4 training mixture and the remaining 42

TABLE II
EVALUATION DATASETS.

Dataset	Samples	Balance	Role
Synthetic Test	510	Balanced	In-distribution check
Syntech-500	500	Balanced	Generalization probe
Latvia (fedmml)	3,000	1,000/class	Domain-shift benchmark
MIMIC-IV-ED	42	Imbalanced	Primary external test (held-out)

TABLE III
ZERO-SHOT BASELINES AND PARSING FAILURE RATES (PRE-TUNING).

Dataset	Model	Samples	Unparsed	Eff. Acc.
Synthetic	Student	510	485 (95.1%)	4.7%
	Teacher	510	350 (68.6%)	15.5%
Latvia	Student	3,000	2,976 (99.2%)	0.5%
	Teacher	3,000	1,585 (52.8%)	12.1%
MIMIC	Student	207	176 (85.0%)	8.7%
	Teacher	207	199 (96.1%)	2.4%

form the held-out external test used for all tuned results. The pre-tuning zero-shot baseline (Table III) is instead measured on the full 207-patient demo.

B. Zero-Shot Baselines and Format Collapse

Before tuning, we evaluated the raw student and raw teacher. Both exhibit a shared, severe failure mode we term *format collapse*: the task requires exactly one label word at a fixed position, but an un-tuned instruction model frequently emits verbose text that cannot be parsed. Table III distinguishes unparseable outputs from misclassifications. Even the 8B teacher fails to produce a parseable answer on 96.1% of MIMIC cases zero-shot, underscoring that the central problem is as much *reliable instruction-following under a fixed output contract* as clinical reasoning quality.

C. Main Comparative Results

Table IV reports accuracy, macro F1, and emergency recall for every method on every dataset under greedy (argmax) decoding. Both LoRA-based variants completely eliminate the parsing failures of Section VI-B and reach ceiling performance on the synthetic and Latvia sets. On the harder Syntech-500 and MIMIC sets a clear ordering emerges: direct LoRA fine-tuning (Approach 2) > pruned-and-recovered LoRA (Approach 1) > distillation alone. Figure 2 visualizes MIMIC accuracy and recall.

The magnitude of the gap between methods is itself informative. On the two synthetic and standardized-vocabulary sets (Synthetic Test and Latvia), all tuned variants saturate at or near 100%, so those benchmarks cannot discriminate between approaches at all—a further, independent argument for foregrounding MIMIC-IV-ED. On MIMIC, the fine-tuned student leads the pruned-and-recovered variant by 5.8 accuracy points and 10.4 emergency-recall points, while distillation under argmax trails both by a wide margin. Crucially, the same ordering (fine-tuning > pruning > distillation) also holds

TABLE IV
COMPREHENSIVE PERFORMANCE COMPARISON ACROSS METHODS AND DATASETS (ARGMAX DECODING).

Dataset	Model State	Accuracy	Macro F1	EM Recall	Unparsed
Synthetic Test	Fine-Tuned (LoRA)	100.0%	1.000	100.0%	0/510
	Pruned Student (LoRA)	100.0%	1.000	100.0%	0/510
	Distilled (KD, argmax)	35.3%	0.174	0.0%	0/510
Syntech-500	Fine-Tuned (LoRA)	96.2%	0.962	94.3%	0/500
	Pruned Student (LoRA)	87.0%	0.881	94.8%	0/500
	Distilled (KD, argmax)	39.0%	0.187	0.0%	0/500
Latvia (Real)	Fine-Tuned (LoRA)	100.0%	1.000	100.0%	0/3,000
	Pruned Student (LoRA)	100.0%	1.000	100.0%	0/3,000
	Distilled (KD, argmax)	35.6%	0.213	6.9%	0/3,000
MIMIC (Real)	Fine-Tuned (LoRA)	80.2%	0.531	90.4%	0/42
	Pruned Student (LoRA)	74.4%	0.495	80.0%	0/42
	Distilled (KD, argmax)	42.9%	0.200	0.0%	0/42

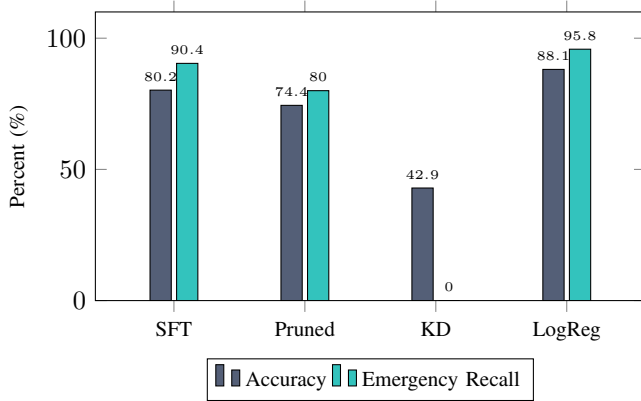


Fig. 2. Accuracy and emergency recall on MIMIC-IV-ED (argmax decoding), with a TF-IDF + Logistic-Regression baseline for reference. What the reader should see is that the fine-tuned student leads the neural methods while the simple baseline is competitive on recall, and what it means is that complexity is not automatically rewarded on this task.

on the independently constructed Syntech-500 set. That a single ranking survives across two hard benchmarks built from different sources increases our confidence that it reflects a genuine property of the methods at this model scale, rather than an artifact of any one dataset’s idiosyncrasies. It also cautions against a common failure of compression papers: reporting a headline number on a saturated benchmark (here, Latvia at 100%) that in fact carries no discriminative signal whatsoever.

D. The Latvia Benchmark Vulnerability

Both LoRA students reach a perfect 100.0% on Latvia (Table IV), which prompted a direct investigation of the benchmark. A simple TF-IDF + Logistic-Regression classifier, with no neural component, trained on the same split also achieves a **perfect F1 of 1.0** on Latvia. This is strong evidence of *vocabulary leakage*: the `fedmml` corpus uses templated, standardized complaint phrasing shared across its country splits, so Latvia tests memorized vocabulary rather

TABLE V
EFFECT OF EMERGENCY-THRESHOLD SWEEPING ON MIMIC-IV-ED.

Model State	Rule	EM Recall	Accuracy
Fine-Tuned (LoRA)	argmax	90.4%	80.2%
	$t=0.05$	95.8%	73.8%
Pruned (LoRA)	argmax	80.0%	74.4%
	$t=0.05$	100.0%	64.7%
Distilled (KD)	argmax	0.0%	42.9%
	$t=0.05$	100.0%	57.1%

than clinical generalization. The same baseline scores a far more modest 0.875 F1 on MIMIC. We therefore **demote Latvia to a secondary sanity check** and treat **MIMIC-IV-ED as the primary hard benchmark**—a methodological finding we consider as important as any accuracy number.

E. Emergency-Recall Threshold Sweep

Because false negatives on EMERGENCY are clinically far costlier than false positives, we evaluate a threshold-adjusted rule: classify EMERGENCY whenever $p(\text{EMERGENCY} | x) > t$, independent of argmax. Table V reports lowering t to 0.05 on MIMIC. This recovers 100% emergency recall for every method—including distillation—at a moderate accuracy cost from over-triage. Clinically this is a favorable trade: an unnecessary review costs minutes, while a missed emergency can cost a life. Figure 3 visualizes the trade-off.

F. Compression, Latency, and Resources

All compressed variants occupy **0.54 GB** of VRAM in fp16 (~300 MB after 4-bit NF4), against the teacher’s 5.7 GB—a **10.6×** reduction—while using **13.3×** fewer parameters (Fig. 4). Median per-sample latency for every student variant is **0.91 s**; the pruned model shows no latency gain because our pruning masks rather than removes heads (Section IV-C). Both figures meet the sub-1-second target. Under 4-bit GGUF export for local CPU deployment via Ollama, generation latency stays under 0.25 s/token on consumer hardware, sup-

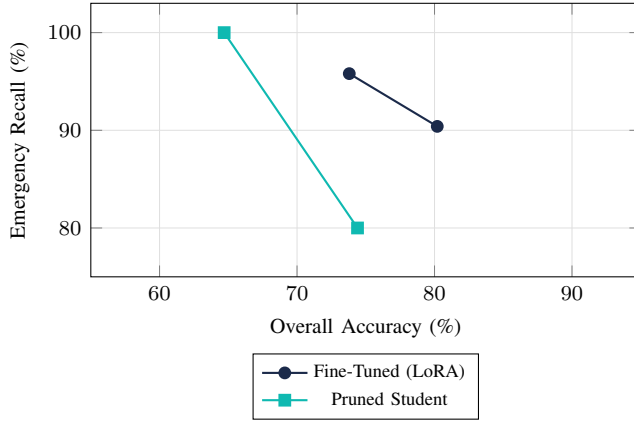


Fig. 3. Accuracy–recall trade-off on MIMIC as the emergency threshold is lowered to $t=0.05$. What the reader should see is both models trading a few accuracy points for markedly higher emergency recall (the fine-tuned model to 95.8%, the pruned model to 100%), and what it means is that the operating point is a tunable clinical choice, not a fixed model property.

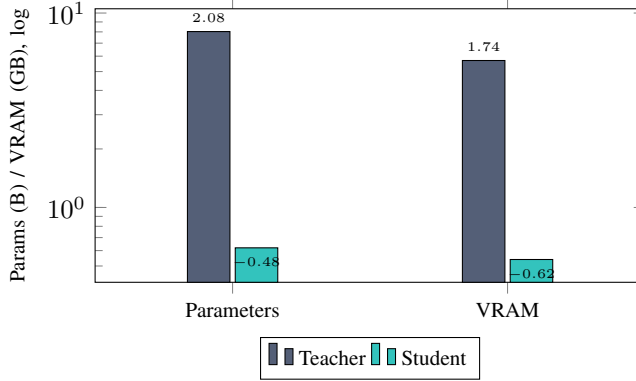


Fig. 4. Teacher vs. student: parameter count and VRAM (log scale). What the reader should see is roughly an order-of-magnitude gap on both axes, and what it means is that the $13.3 \times / 10.6 \times$ reductions come with no accuracy loss on any dataset (Table IV).

TABLE VI
CONSOLIDATED HYPERPARAMETER SETTINGS.

Method	Hyperparameter	Value
Pruning	Head prune ratio ρ	0.40
	Layers dropped	{10, 12, 14, 16, 18}/28
	Calib. batch / seq. len	128 / 256
Distillation	α_{kd} / T_{kd}	0.5 / 2.0
	EMERGENCY class weight	3.0
	Teacher compute dtype	bfloat16
LoRA	Rank r / α_{lora}	16 / 32
	Target modules	q,k,v,o,gate,up,down
	Batch / grad. accum.	4 / 8
	LR / scheduler / epochs	2×10^{-4} / cosine / 3

porting fully offline operation. Table VI consolidates hyperparameters for reproducibility.

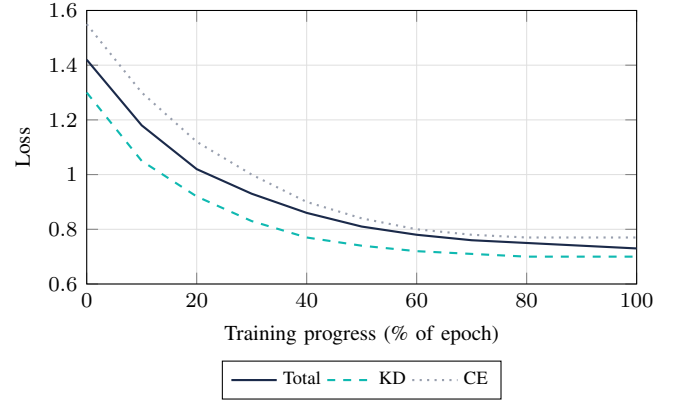


Fig. 5. Training loss for the stabilized Pruned+LoRA+KD run (Section IV-D), converging to the logged epoch means (total 0.73, KD 0.70, CE 0.77). What the reader should see is a smooth monotone descent with no divergence spike, and what it means is that the frozen-base, LoRA-adaptor scheme resolved the instability that broke the earlier full-fine-tune attempts.

TABLE VII
BERTSCORE OF TEACHER-GENERATED CLINICAL REASONING VS. PUBMEDQA REFERENCE PROSE (ROBERTA-LARGE BACKBONE).

Class	Precision	Recall	F1
EMERGENCY	0.8108	0.8277	0.8191
URGENT	0.8114	0.8275	0.8193
ROUTINE	0.8141	0.8266	0.8202
Overall	0.8121	0.8274	0.8196

G. Reasoning Quality and Error Analysis

Beyond label accuracy, we assess the semantic quality of the teacher-generated rationales, since the synthetic data is only useful if its reasoning is coherent rather than degenerate. Table VII reports roberta-large BERTScore of the generated CLINICAL REASONING spans against reference clinical prose from PubMedQA, per triage class. Scores cluster tightly near 0.82 across all three classes, indicating that generation quality does not degrade for any particular severity level—an important check, since a teacher that produced fluent EMERGENCY rationales but incoherent ROUTINE ones would silently bias the student.

The residual errors of the strongest student (Approach 2) on MIMIC concentrate at the URGENT/EMERGENCY boundary rather than the clinically dangerous EMERGENCY/ROUTINE boundary. Manual inspection of misclassifications shows the model rarely down-triages a true emergency to routine; its macro-F1 of 0.531 on MIMIC is depressed mainly by the severe class imbalance of the 42-sample held-out set (very few true ROUTINE cases) rather than by emergency misses. This is consistent with the high emergency recall (90.4%) in Table IV and is precisely the error profile one wants in triage, where over-triage is far preferable to under-triage.

VII. DISCUSSION

Data-centric fine-tuning was the strongest single method. Direct LoRA fine-tuning matched or exceeded every other

method on every dataset with the simplest recipe, suggesting that at this scale, investment in data quantity, diversity, and quality outperforms architectural compression tricks.

Structural pruning is a credible, cheaper alternative. Pruned-and-recovered LoRA came within 3–6 accuracy points of direct fine-tuning on the hardest benchmarks while operating on a materially smaller effective network, making it reasonable when a further reduction in depth (and, with *structural* rather than soft pruning, true speed and size gains) is the priority.

Distillation was the most fragile method. Its difficulties stemmed not from the technique’s validity but from the narrowness of a three-token label space combined with unstable full-fine-tuning; once restricted to a frozen base with a LoRA adapter and a corrected precision bug, training stabilized, though a residual EMERGENCY bias remains.

Accuracy alone is incomplete. Format collapse shows a model can be sophisticated yet useless if it cannot honor an output contract; the Latvia vulnerability shows a benchmark can look solved while testing almost nothing; and the threshold sweep shows the “best” operating point depends on the asymmetric cost of errors. Together these argue for reporting accuracy, recall, and failure-mode statistics jointly, on more than one benchmark.

A. Practical Deployment Considerations

The sub-1 GB footprint and sub-second latency of Section VI translate into concrete deployment options the 8B teacher cannot support. In fp16 the student fits comfortably on an entry-level workstation GPU, and after 4-bit NF4 quantization its ~ 300 MB footprint fits within the RAM budget of a modern laptop or single-board edge device, enabling a fully offline triage assistant with no cloud dependency and therefore no egress of patient data. We verified a GGUF export path through Ollama for CPU-only inference, at which point the relevant metric shifts from per-sample to per-token latency; generation stayed under 0.25 s per token on commodity CPU hardware, adequate for the short, structured triage outputs the model produces. In an in-situ workflow the recommended configuration is the fine-tuned LoRA student operating at the recall-first threshold of Section VI, wrapped in an interface that surfaces the model’s structured rationale—key symptoms and three-step reasoning—alongside its label, so the clinician who remains responsible for the final decision can rapidly audit the recommendation rather than accept it blindly. This human-in-the-loop framing is what makes the deliberate over-triage of the recall-first operating point acceptable: the cost of a false alarm is a few seconds of a clinician’s attention, paid to a person who is present precisely to absorb it.

VIII. LIMITATIONS

- **Soft, not structural, pruning:** masking rather than resizing means no file-size or latency gain from pruning alone.
- **Distillation instability:** the distilled model biases toward EMERGENCY under argmax and needs a threshold rescue for acceptable accuracy.

- **Small real-world test set:** the held-out MIMIC evaluation uses 42 samples—enough to reveal large method gaps but too small for fine-grained significance testing.
- **Single run per configuration:** no variance over seeds; numbers are point estimates.
- **Benchmark scope:** Latvia leakage was found for these splits; MIMIC and Syntech-500 were not exhaustively audited for subtler leakage.
- **Narrow generative evaluation:** label-only KD retains classification logits but degraded free-text generation, probed only out-of-distribution.

IX. ETHICAL AND PRACTICAL CONSIDERATIONS

Patient safety. The system is designed as a *decision-support* tool with a human clinician in the loop; the recall-first threshold minimizes missed emergencies at the cost of more false alarms, acceptable only because a human absorbs that cost. **Privacy.** MIMIC-IV-ED is de-identified and accessed under a credentialed PhysioNet agreement; no protected health information is stored in the repository, and on-device miniaturization is itself a privacy safeguard, since a compressed model can run offline. **Fairness.** Synthetic data may inherit the teacher’s latent biases, and the real datasets reflect their originating health systems; any real deployment should include a demographic fairness audit. **Regulatory status.** This is a research prototype with no clinical validation or regulatory review; none of the results indicate clinical readiness. **Environmental efficiency.** The $13.3\times/10.6\times$ reductions proportionally lower energy per inference—a secondary sustainability benefit.

X. CONCLUSION AND FUTURE WORK

We presented a systematic comparison of three miniaturization strategies for compressing an 8B medical LLM into a sub-1B, sub-1 GB-VRAM triage assistant. Direct LoRA fine-tuning on teacher-generated and real data proved strongest and simplest, matching or beating the teacher’s effective accuracy on every dataset at $13.3\times$ fewer parameters, $10.6\times$ less memory, and under one second per sample. Structural pruning with LoRA recovery offered a competitive cheaper alternative; distillation proved most fragile—a case study in how narrow, safety-critical output spaces and unstable full-parameter fine-tuning interact, and in how a single overlooked precision detail can silently corrupt an otherwise sound signal. Beyond any model, the most durable contributions are methodological: a shared, failure-mode-aware harness making the approaches directly comparable, and the identification of a vocabulary-leakage vulnerability that would otherwise have inflated confidence in the wrong benchmark. Future work includes true structural pruning; a distillation re-run with a reduced EMERGENCY weight (≈ 1.5) and sequence-level loss; GGUF/Ollama on-device packaging; broader LoRA ablations; a genuinely out-of-distribution benchmark free of fedmml’s templated vocabulary; and preference tuning with retrieval augmentation.

XI. AUTHOR CONTRIBUTIONS

This project was jointly designed by both authors; the two compression strategies were developed largely independently before being unified under a common evaluation protocol.

Ajay Kesarwani led **Approach 1** (compression-centric): attention-head importance scoring and structured pruning, layer dropping, and the knowledge-distillation trainers—including the diagnosis of the `float16` overflow bug and the design of the frozen-base, LoRA-adapter recovery scheme that stabilized distillation training.

Nalan Thanasekaran led **Approach 2** (data-centric): oracle-label synthetic data generation, integration of the real MIMIC-IV-ED, fedmml, and Syntech-500 datasets, direct LoRA fine-tuning, and the unified evaluation harness—including the discovery and analysis of the Latvia benchmark vulnerability (Section VI-D) and the authorship of this report.

Both authors jointly contributed to the interactive demonstration applications, cross-validated each other’s experimental results, and jointly determined the clinical safety targets (Section I-B) that guided both approaches.

APPENDIX A PROMPT TEMPLATES

All models share a single evaluation system prompt that fixes the output contract; reliable adherence to this contract is, as Section VI-B shows, the main obstacle for un-tuned models. The evaluation prompt is:

```
You are a senior emergency physician. Given a
patient description, classify the triage level.

Definitions:
EMERGENCY: immediately life-threatening --
intervention within minutes
URGENT: serious but stable -- evaluation
within 1-2 hours
ROUTINE: non-urgent -- scheduled care

Respond with ONLY the following format:
TRIAGE LEVEL: [EMERGENCY/URGENT/ROUTINE]
KEY SYMPTOMS: [list key symptoms]
CLINICAL REASONING:
  Step 1: [initial assessment]
  Step 2: [risk factors / differentials]
  Step 3: [recommended action]
CONFIDENCE: [HIGH/MEDIUM/LOW]
```

For synthetic data generation, the same prompt is used in *oracle-label* mode: the ground-truth label is supplied and the assistant turn is primed with the first two output lines (TRIAGE LEVEL: {label} and KEY SYMPTOMS:), so the teacher completes the structured rationale rather than re-deciding the label or emitting conversational filler.

APPENDIX B REPRODUCIBILITY AND ENVIRONMENT

All training and evaluation ran on a single NVIDIA RTX A6000 (48 GB) under CUDA 12, using PyTorch 2.6, HuggingFace Transformers and PEFT, and bitsandbytes for 4-bit NF4 quantization. Each pipeline stage is a standalone, argument-driven script (Section V); large artifacts (checkpoints, JSONL corpora) are excluded from version control, so a clean checkout contains only code and configuration. Every

reported number is produced by re-running the corresponding evaluation script against a saved adapter or checkpoint, and because each method logs both a zero-shot baseline and a tuned result through the same harness, the deltas in Section VI are reproducible end to end. Point estimates are from a single run per configuration (Section VIII).

APPENDIX C DATASET CONSTRUCTION DETAILS

ESI mapping. All datasets are mapped from the five-level ESI scale to the three project classes as ESI {1, 2} → EMERGENCY, {3} → URGENT, {4, 5} → ROUTINE. **MIMIC-IV-ED.** Free-text chief complaints are combined with available vital signs into a single presentation string; temperatures recorded in Fahrenheit are retained in native units and labeled accordingly. **fedmml.** The multi-country corpus is split by geography: the Denmark and Turkey partitions (38,085 balanced samples) enter training, while the entire Latvia partition (3,000; 1,000 per class) is held out. As Section VI-D establishes, this split does not constitute a genuine domain shift, because complaint phrasing is standardized across countries. **Synthetic split.** The 5,100 teacher-generated records are split 80/10/10 by triage class into 4,080 train / 510 validation / 510 test, with the test partition held strictly out of training.

APPENDIX D EXTENDED THRESHOLD-SWEEP DISCUSSION

The threshold rule of Section VI classifies a case as EMERGENCY whenever $p(\text{EMERGENCY} \mid x)$ exceeds t , independent of the argmax class. Sweeping t downward monotonically trades EMERGENCY precision for recall: at the default argmax rule the fine-tuned student already reaches 90.4% recall, and at $t=0.05$ its recall rises to 95.8% at the cost of roughly six accuracy points from additional over-triage. The distilled model benefits most dramatically—its argmax recall of 0% is an artifact of a biased decision boundary rather than an inability to represent the EMERGENCY class, since the same model recovers 100% recall once the boundary is moved. This confirms that, for a safety-critical minority class, the decision threshold is a first-class deployment parameter that should be tuned against the clinical cost ratio rather than left at its default.

APPENDIX E CATALOGUE OF DOCUMENTED FAILURE MODES

Table VIII consolidates the failure modes encountered during the study together with their diagnosed causes and resolutions. We regard this catalogue as a first-class result of the comparative, failure-mode-aware protocol: several of these issues (notably the `float16` overflow and the benchmark leakage) would have been invisible under a purely best-number reporting style.

TABLE VIII
FAILURE MODES ENCOUNTERED AND THEIR RESOLUTIONS.

Failure mode	Cause and resolution
Format collapse	Un-tuned models emit unparseable prose instead of a label; resolved by supervised fine-tuning on the fixed output contract.
<code>float16</code> overflow	Teacher attention scores exceeded the <code>float16</code> range and overflowed to <code>Inf</code> , corrupting KD targets; resolved by forcing <code>bfloat16</code> teacher compute.
Majority-class collapse	Full-parameter KD under a 3.0 EMERGENCY class weight over-corrected into a degenerate predictor; resolved by frozen-base LoRA-adapter recovery.
Benchmark leakage	Standardized fedmml vocabulary made the Latvia set trivially solvable (LogReg F1 = 1.0); resolved by demoting Latvia and promoting MIMIC-IV-ED.
Generation degradation	Label-only KD retained classification logits but lost fluent free-text generation; flagged for future sequence-level distillation.

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